

Multimodal Fine-Tuning

LoRA, QLoRA, DPO, hallucination reduction, and catastrophic forgetting in vision-language models

10 min

Duration

B3 – Apply

Bloom's

L3 – Intermediate

Level

TH-000012

Prerequisite

Session Overview

From general-purpose to domain-specific VLMs — the practical guide to multimodal fine-tuning

What You Will Learn

- Why pre-trained VLMs fail on domain-specific data
- Full fine-tuning vs. parameter-efficient fine-tuning (PEFT)
- LoRA: low-rank adaptation and the weight update formula
- QLoRA: fine-tuning a 13B VLM on a single 24GB GPU
- Multimodal instruction tuning: (image, instruction, response) triplets
- RLHF vs. DPO for aligning multimodal models
- Catastrophic forgetting and five mitigation strategies
- Hallucination reduction via fine-tuning

Concept at a Glance

- Duration: 10 minutes
- Bloom's Level: B3 – Apply
- Difficulty: L3 – Intermediate
- Module: 4.3 Multimodal Fusion and Training
- Prerequisites: Training Multimodal Systems (TH-000012)
- Completes Module 4.3

Learning Objectives

By the end of this session you will be able to:

10 min

Duration

B3 – Apply

Bloom's

L3 – Intermediate

Level

TH-000012

Prerequisite

- 1 Distinguish full fine-tuning from parameter-efficient fine-tuning in the multimodal context
- 2 Apply LoRA to vision-language models and identify which weight matrices to target
- 3 Explain QLoRA's 4-bit quantisation trick and calculate the resulting GPU memory savings
- 4 Construct a multimodal instruction-tuning dataset for a specific domain
- 5 Compare RLHF and DPO for aligning multimodal models with human preferences
- 6 Identify catastrophic forgetting and describe five mitigation strategies

SECTION 1 — WHY FINE-TUNE?

The Domain Gap Problem

General-purpose VLMs fail on domain-specific visual distributions and specialised vocabularies

Why Pre-trained VLMs Need Fine-Tuning

LLaVA, BLIP-2, GPT-4V are general-purpose — they perform well on natural images but fail on domain-specific data

Medical Imaging

✗ Cannot read chest X-rays, CT scans, or MRIs — DICOM format and clinical vocabulary absent from natural image training data. 'Pneumothorax' is not in WebCap.

✓ Fine-tune on PMC-15M (15M biomedical figure-caption pairs from PubMed). Achieves clinical-level VQA on PathVQA and VQA-RAD benchmarks.

Industrial Inspection

✗ Cannot detect micro-cracks in circuit boards, weld defects, or surface anomalies — domain-specific visual patterns never appear in web-scraped data.

✓ Fine-tune with LoRA on domain images + (defect, description) pairs. Preserves general knowledge via LoRA isolation, adds visual specialisation.

Scientific Figures

✗ Misinterprets chemical structures, biological diagrams, and research plots as generic images. Cannot parse axis labels or understand figure captions in context.

✓ ScienceQA dataset fine-tuning. Include chart QA (ChartQA), document understanding (DocVQA), and mathematical reasoning (MathVista) tasks.

Legal/Financial Documents

✗ No exposure to contract language, table structures in financial reports, or signature verification tasks during general pre-training. Hallucination rate is very high on structured documents.

✓ Instruction tuning on domain-specific (image, question, answer) triplets generated by GPT-4o from real documents. Often requires human review of synthetic answers.

ROI: Domain fine-tuning routinely reduces task error rates by 30–60% vs out-of-box model. Even 1,000 domain-specific (image, QA) pairs can yield significant improvement. The cost is often <\$50 for LoRA fine-tuning.

SECTION 2 — LORA AND QLORA

Parameter-Efficient Fine-Tuning

Fine-tune only 1–3% of parameters while retaining 98%+ of the model's general capability

LoRA: Low-Rank Adaptation

Instead of updating W directly, inject small low-rank matrices A and B — train only those

$$W' = W + \Delta W = W + (\alpha/r) \cdot B \cdot A \quad \text{where } A \in \mathbb{R}^{(d \times r)}, B \in \mathbb{R}^{(r \times d)}, r \ll d$$

W = frozen original weight | r = rank (typically 4–64) | α = scaling factor | $B \cdot A$ = learned update

Why low-rank?:

Pre-training updates W have low intrinsic dimensionality. The effective rank of fine-tuning updates is empirically much smaller than d . A 4096×4096 matrix (16M params) is well approximated by rank-16 BA (128K params) — 99.2% reduction.

Where to apply in a VLM?:

LLM attention layers (Q, K, V, output projection) — ALWAYS. The MLP projector — train fully (it's small). ViT layers — optionally, for domain-specific visual features (medical imaging, satellite imagery).

Practical numbers (7B LLaVA, $r=16$):

Full fine-tuning: 7B trainable parameters. LoRA (LLM Q,K,V,O): ~80M trainable parameters — 98.9% reduction. Performance gap: typically <2% on standard benchmarks. Training memory: 40GB → 20GB.

Key implementation detail:

Initialise $B=0$ so $\Delta W=0$ at start — fine-tuning begins from the pretrained model without disruption. A is randomly initialised from a normal distribution. The (α/r) scaling prevents the learning rate from needing adjustment.

Merge at inference: LoRA adapters can be merged into the base model ($W' = W + BA \cdot \alpha/r$) before deployment. The merged model has identical inference latency to the original — zero overhead. This makes LoRA the dominant production PEFT method.

QLoRA: Fine-Tuning a 13B VLM on a Single 24GB GPU

Dettmers et al. (2023) — 4-bit NormalFloat quantisation + LoRA adapters = 75% GPU memory reduction

QLoRA keeps the base model frozen and quantised to 4-bit NF4 (NormalFloat4 — optimal for normally distributed weights). LoRA adapters are trained in bfloat16 precision. Double quantisation (quantise the quantisation constants) saves additional 0.5 GB per billion parameters.

Strategy	% Params Trained	GPU Memory	Forgetting Risk	Best For
Full Fine-Tuning	100%	60–160GB	High	Large data (>100K), major domain shift
Projector Only	<1%	12–20GB	Very Low	Quick task adaptation, same domain
LoRA (LLM only)	1–3%	20–40GB	Low	Most production cases — best default
QLoRA (LLM)	1–3%	12–24GB	Low	Single GPU, budget-constrained — consumer hardware
LoRA (full VLM)	3–8%	24–48GB	Medium	Domain-specific visual features (medical, satellite)

QLoRA numbers for LLaVA-13B (r=16): Full fine-tuning: 4×80GB A100 (~\$500/epoch). QLoRA: 1×24GB RTX 3090 (~\$30/epoch). Performance difference: typically <2% on standard VQA benchmarks. Democratises multimodal fine-tuning for academic labs and startups.

SECTION 3 — ALIGNMENT AND HALLUCINATION

DPO Alignment and Catastrophic Forgetting

Making VLMs helpful and honest — and keeping them from forgetting everything they knew

DPO vs RLHF for Multimodal Alignment

Both methods align VLMs with human preferences — DPO is simpler, more stable, and now the preferred approach

RLHF (3-stage, complex)

Stage 1: Collect human preference pairs

- Show annotators two VLM responses to same image+question
- They choose the better one

Stage 2: Train a reward model

- A separate VLM that scores (image, question, response) triples
- Based on collected preferences (50K–500K comparisons)

Stage 3: PPO fine-tuning

- Update VLM to maximise reward model scores

Risks:

- Reward hacking: model learns to fool the reward model
- Hallucination can increase — RLHF rewards fluency, which incentivises confident hallucination
- Very expensive: requires 3 models simultaneously

DPO — Direct Preference Optimisation ✓

Formula:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}[\log \sigma(\beta \log(\pi_{\theta}(y_w|x)/\pi_{\text{ref}}(y_w|x)) - \beta \log(\pi_{\theta}(y_l|x)/\pi_{\text{ref}}(y_l|x)))]$$

y_w = preferred response, y_l = rejected response

Key advantages:

- No reward model needed
- No PPO training loop
- Direct supervised learning on preference pairs
- More stable training

Used in:

- LLaVA-RLHF, InstructBLIP-DPO, RLHF-V
- RLHF-V (2024): segment-level DPO corrections → SOTA hallucination reduction in VLMs

Preference: DPO is now the preferred alignment method for multimodal models. Simpler, faster, and achieves comparable or better hallucination reduction than RLHF. The trend in 2024: use synthetic preference data generated by GPT-4o to avoid expensive human annotation.

Catastrophic Forgetting and Hallucination Reduction

Two critical failure modes in multimodal fine-tuning — and five proven mitigation strategies

Catastrophic Forgetting: When fine-tuned on domain-specific data, a VLM's general capabilities (everyday photo understanding, commonsense QA) degrade. A model fine-tuned only on medical images may no longer answer questions about ordinary photographs correctly.

1

Replay Buffer: Mix 10–20% general data (LLaVA-Instruct) with domain data during fine-tuning. The simplest and most effective mitigation. Standard practice.

2

LoRA Isolation: Keep base model FROZEN; only LoRA adapters change. The base model retains all general knowledge. Domain-specific knowledge lives entirely in adapters.

3

EWC (Elastic Weight Consolidation): Add a regularisation term that penalises large changes to weights identified as important for general capabilities. Computationally expensive but powerful.

4

Multi-task Fine-tuning: Include diverse tasks (general VQA, captioning) alongside domain data during fine-tuning. Maintains breadth while adding depth.

5

Progressive Training: Train on general data first; domain data second with a lower learning rate. Treats domain adaptation as incremental rather than abrupt.

Hallucination reduction: Include negative examples ('what is NOT in this image?') in instruction tuning. Apply DPO/RLHF-V with segment-level corrections. Add bounding box prediction objectives to force precise visual grounding. Use POPE benchmark to measure hallucination rate during training.

Tools and Frameworks for Multimodal Fine-Tuning

Production-ready frameworks that make LoRA/QLoRA fine-tuning of VLMs accessible on consumer hardware

Framework	Key Features	Best For
LLaVA GitHub (haotian-liu/LLaVA)	Reference LLaVA implementation. Full FT, LoRA, QLoRA. Well-documented scripts.	LLaVA variants; academic reproducibility
HuggingFace PEFT + TRL	peft: LoRA/QLoRA injection. trl: SFT, DPO, RLHF loops. Any HF VLM.	Most flexible; largest model selection
LLaMAFactory	Web UI + CLI for 100+ models including VLMs. LoRA, QLoRA, DPO minimal code.	Non-ML engineers; rapid prototyping
Axolotl	YAML configuration. Strong community for multimodal extensions.	Custom training recipes; reproducibility
DeepSpeed + ZeRO3	Distributed training for large VLMs. ZeRO-3 shards all states across GPUs.	Full fine-tuning of 30B+ VLMs

Recommended stack: HuggingFace PEFT (LoRA injection) + TRL (DPO training) + DeepSpeed ZeRO-2 (memory efficiency) + Weights & Biases (experiment tracking). This combination handles 95% of multimodal fine-tuning use cases from 3B to 70B parameter VLMs.

Key Takeaways

The six essential points from this concept

Fine-tuning closes the domain gap General VLMs fail on medical images, industrial inspection, scientific figures. Domain fine-tuning reduces error by 30–60% even with 1,000 examples.

LoRA: 99% parameter reduction $W' = W + (\alpha/r)BA$. Rank-16 LoRA on a 7B VLM: 7B $\rightarrow$ 80M trainable params. <2% performance gap vs full fine-tuning. Merge at inference for zero latency overhead.

QLoRA: single GPU for 13B VLM 4-bit NF4 quantisation of base model + bfloat16 LoRA adapters. RTX 3090 (24GB) trains LLaVA-13B for ~\$30/epoch vs \$500 on 4xA100.

DPO over RLHF for alignment Direct Preference Optimisation trains on (preferred, rejected) pairs without a reward model or PPO loop. More stable, comparable or better hallucination reduction.

Catastrophic forgetting needs active mitigation Replay buffers (mix 10–20% general data) + LoRA isolation are the two most practical strategies. Never fine-tune on domain data alone without a forgetting mitigation.

Note: Module 4.3 complete — next: AI-MM-IN-CD-000003 Building Multimodal Applications

AI-MM-IN-TH-000013 Complete

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